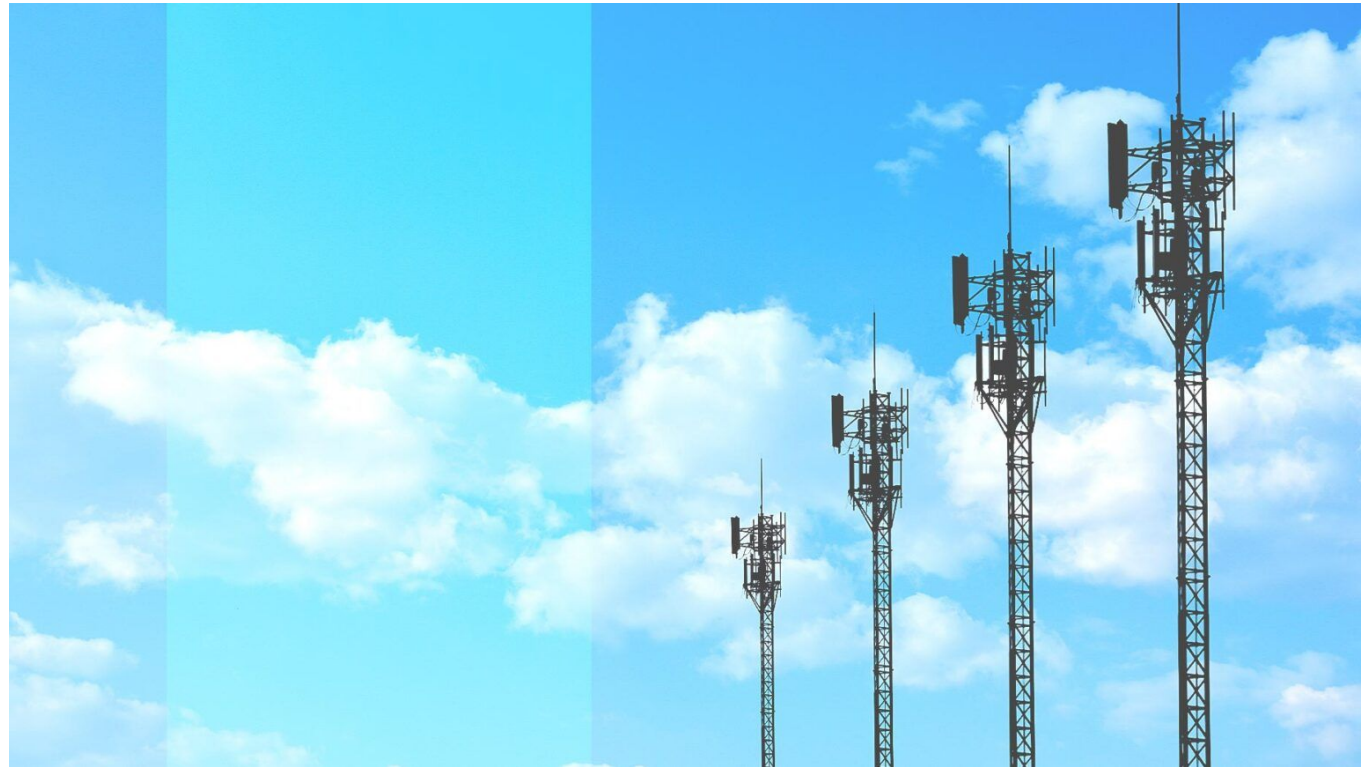


The RFRL Gym:

A testbed for training and evaluating reinforcement learning for
cognitive radio systems

Danny Rosen, Alyse Jones, Dr. Chris
Headley

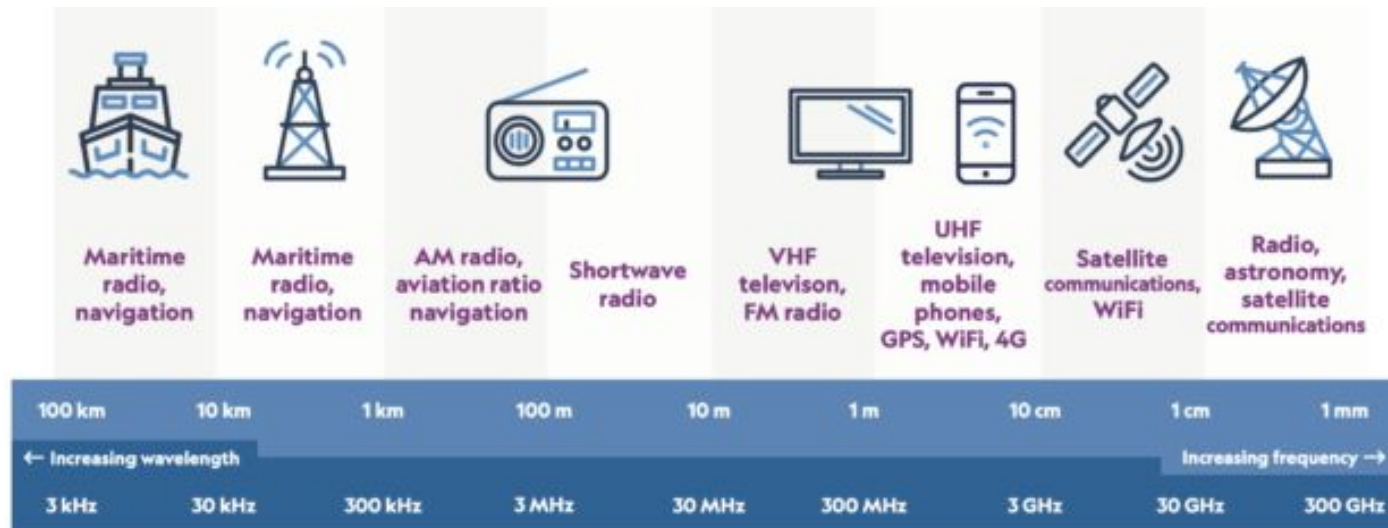
- Introduction to problem space
 - Wireless Spectrum Problems: Interference and congestion
 - DSA and Jamming
- Motivation
- Contributions
- RFRL Gym Framework
- Validation and Analysis
- Future Work and Conclusions



Wireless Spectrum



- Set of electromagnetic frequencies that are used for wireless communication
- Continuous Frequencies grouped into discrete “bands” or “channels”
- Spectrum in our daily lives - Cellular, WIFI, Bluetooth
- Problem: Interference
 - Multiple transmissions at the same frequency
 - Receiver can not correctly interpret correct signal

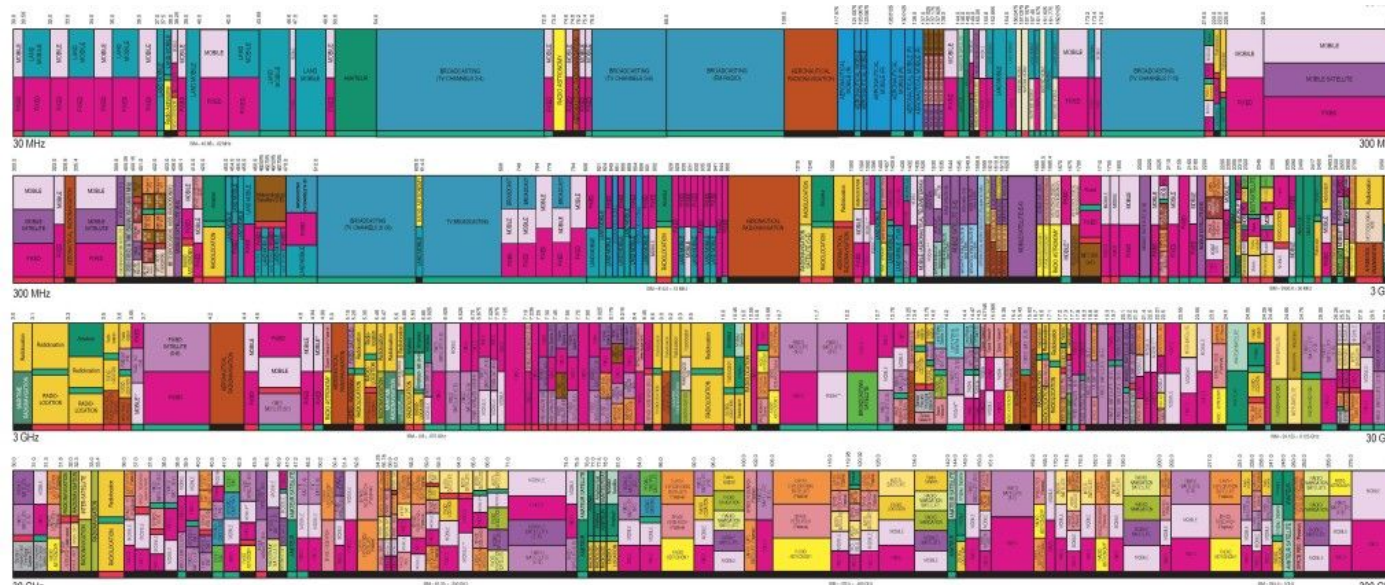


Acquired from
[1]

Causes of Interference



- Military: Electronic Warfare
- Commercial: Interactions between consumer devices
 - 2010-2021: 183x increase in wireless spectrum use [2]
- FCC Spectrum Allocation



Acquired from [3]

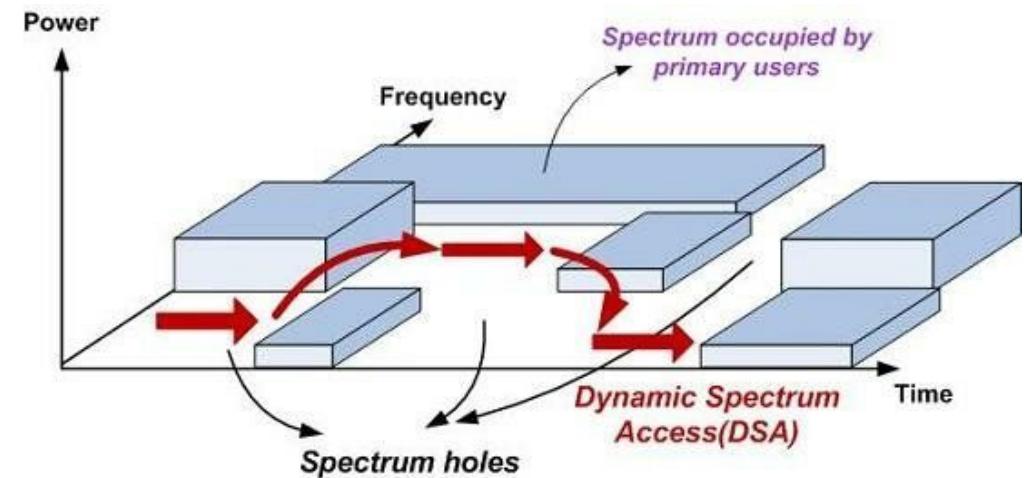
Cognitive Radio (CR)

Radio with the ability to dynamically change transmission parameters (center frequency, bandwidth) based on spectrum sensing results.



Dynamic Spectrum Access (DSA)

Algorithmically switch to vacant frequencies to avoid other entities. Spectrum holes found by Spectrum Sensing.

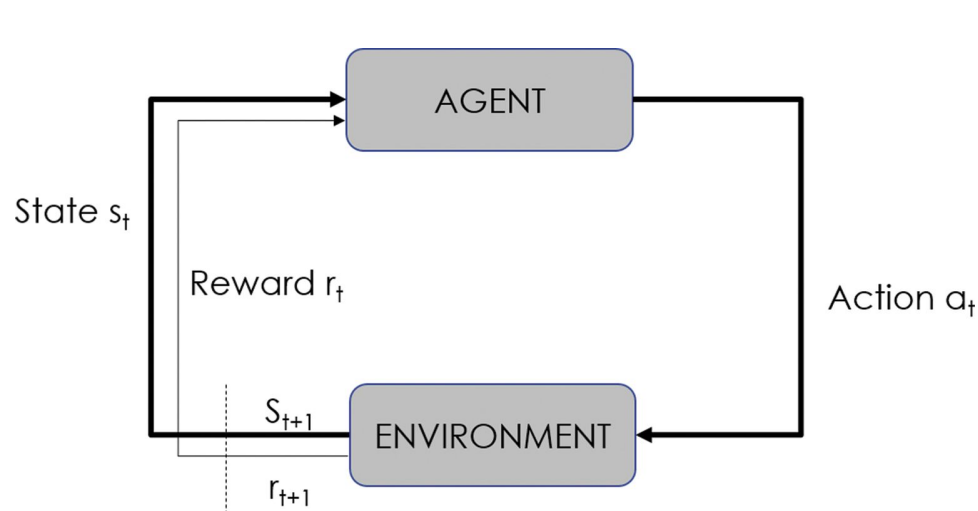


Acquired from [4]

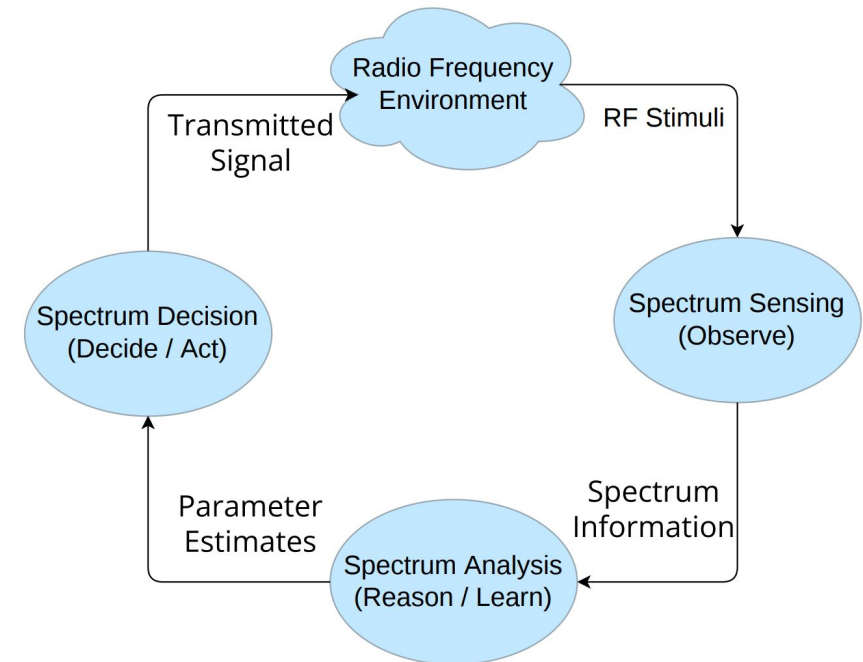
CRs with DSA alone cannot understand where spectrum users will go next, resulting in interference

OODA: Observe Orient Decide Act

Understanding the expected behavior of spectrum activity through **prediction** with Reinforcement Learning



Acquired from [5]



Can use Reinforcement Learning (RL) with CRs to enable **prediction** in its cognition to understand the dynamics and behavior of external factors affecting the RF environment.

1. **Enhance** research capabilities in wireless spectrum applications
 - Easy-to-use cognitive radio-centric research tool
2. **Address** pressing problems of wireless spectrum management
 - Reducing interference in the wireless spectrum
3. **Expand** interest and knowledge in RL-based wireless communications

- **The RFRL Gym** - a customizable tool built in Open AI gymnasium to simulate and test the performance of reinforcement learning (RL) for cognitive radio applications (such as DSA and jamming)
- The RFRL Gym contains the following **novel features**:
 - Integration with Third Party RL packages
 - Customizable Scenarios
 - Out of the box entities
 - Multiple rendering options
 - GUI

Features	RFRL Gym	GrGym [14], [15]	ns3-Gym [17]
Flexible Scenario Design	✓	✓	
RL Package Compatibility	✓	✓	✓
Spectrum Sensing Capabilities	✓		✓
Multi-Agent Capabilities			✓
Ease of Use	✓		
Graphical User Interface	✓		
Hardware Compatible		✓	

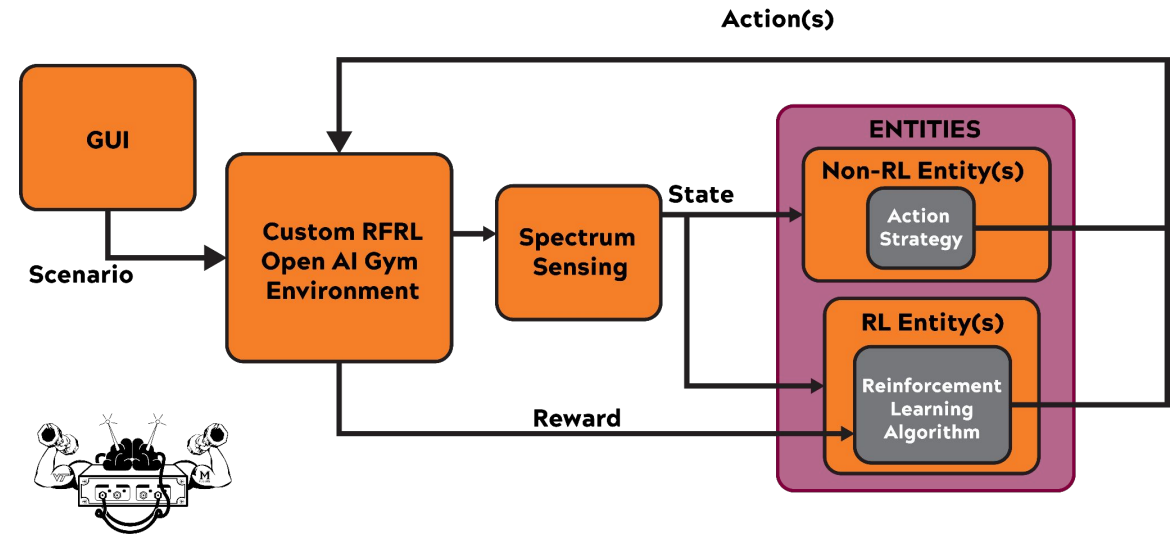


The RFRL Gym Framework

The Development of a Custom OpenAI Gym Environment
for Cognitive Radio Applications

Why OpenAI Gymnasium?

- OpenAI Gymnasium Integration
 - Standardized API
 - State space
 - Action space
- Easy integration with external RL packages
 - Stable Baselines, Hugging Face, Ray RLLib, etc.
 - Pytorch, Tensorflow, etc.

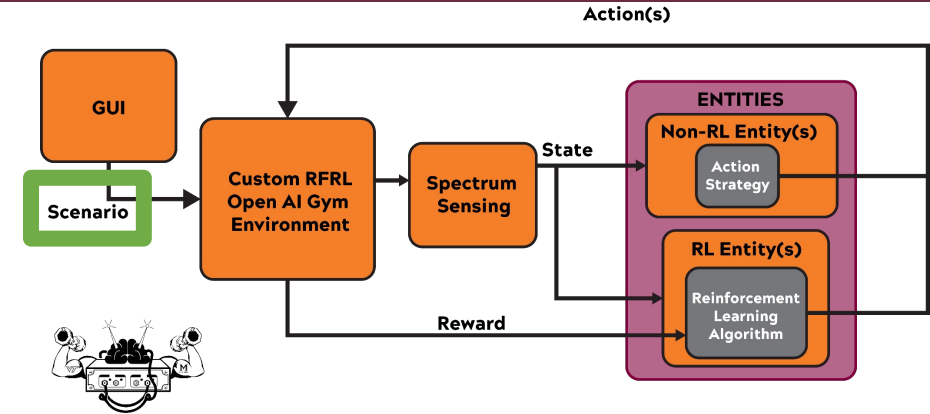


Scenario: Configuration of the simulation

- Entities: Signals moving in spectrum
- Observation modes: Classification and Detection
- Reward modes: Jamming or DSA
- Render Modes: Terminal or PyQt

$$DSA \text{ Reward} = \begin{cases} 1 & \text{no collision} \\ -1 & \text{collision} \end{cases}$$

$$Jamming \text{ Reward} = \begin{cases} 1 & \text{collision} \\ -1 & \text{no collision} \end{cases}$$



Scenario File Generation



Environment

Set The Number Of Channels:

Set The Max Time Step:

Select The Observation Mode:

☐ Classify ☐ Detect

Select The Rewarding Mode:

☐ DSA ☐ Jamming

Entity

Entity Type:

Set The Channels:

Will The Entity Start in an On or Off Position?:

How Many Time Steps Before an Entity Turns Off:

How Many Time Steps Before an Entity Turns On:

What Time Step Does an Entity Start in the Gym:

What Time Step Does an Entity Stop in the Gym:

Select The Hop Type:

☐ Sequential ☐ Random

Scenario Overview

Environment
Render
Entity

Select a Target Entity:

Render

Set The Number Of Frames Per Second:

Set The Number Of Render History:

Select The Render Mode:

☐ Terminal ☐ Matplotlib

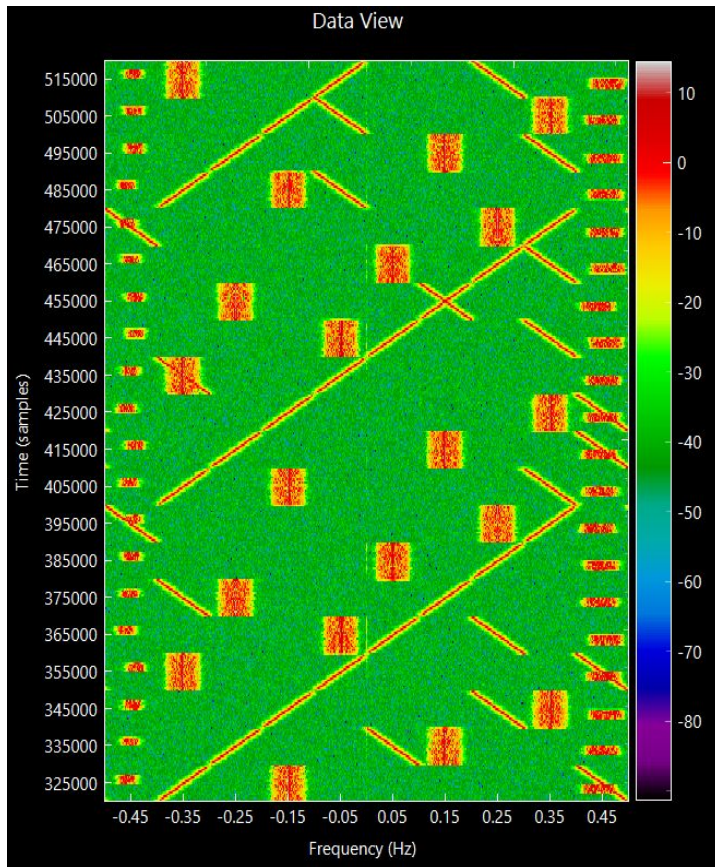
Entity Behavior

I hop around based on the pattern you want~

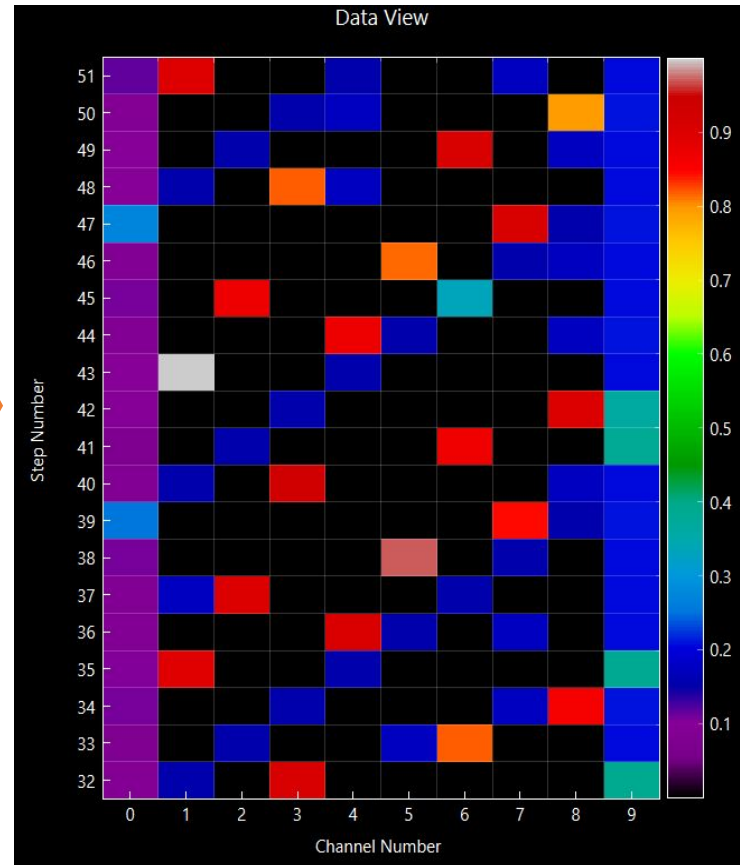
```
{
  "environment":
  {
    "num_channels": 10,
    "max_steps": 100,
    "observation_mode": "detect",
    "reward_mode": "jam",
    "target_entity": "constant_freq_1"
  },
  "entities":
  {
    "constant_freq_1":
    {
      "type": "ConstantFreq",
      "channels": [5],
      "onoff": [1,1,0],
      "modem_params":
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        "type": "qam",
        "order": 16,
        "filter": "RRC",
        "center_frequency": [-0.1,0.1],
        "bandwidth": 0.25,
        "start": 0.25,
        "duration": 0.25
      }
    }
  },
  "render":
  {
    "render_mode": "pyqt",
    "render_fps": 2,
    "render_history": 20,
    "render_background": "black"
  }
}
```

Translating the Wireless Spectrum into the RFRL Gym

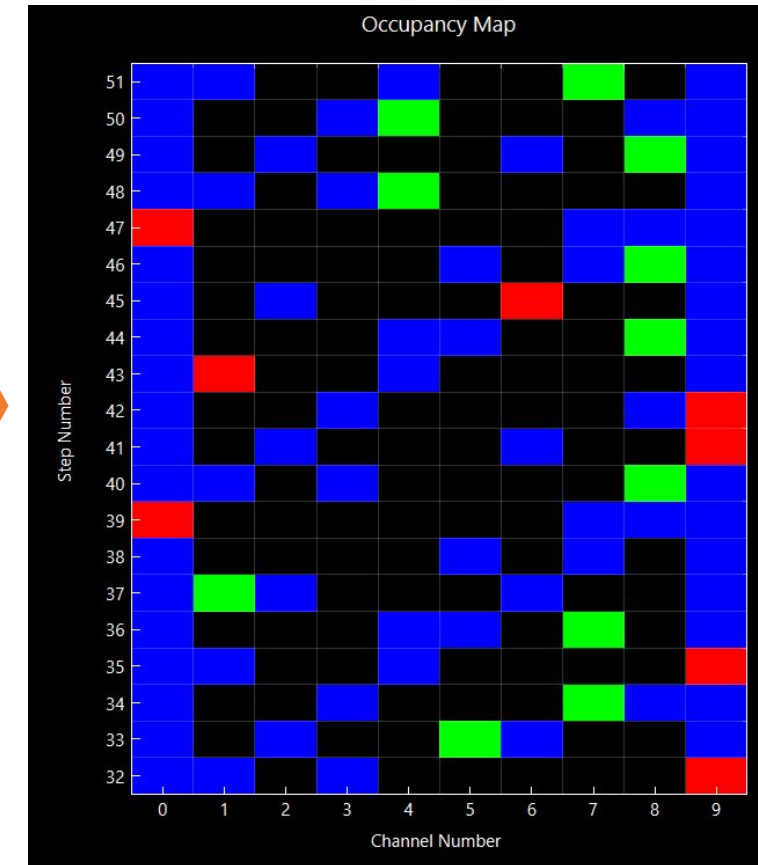
Spectrum View



Sensing View



Occupancy Map

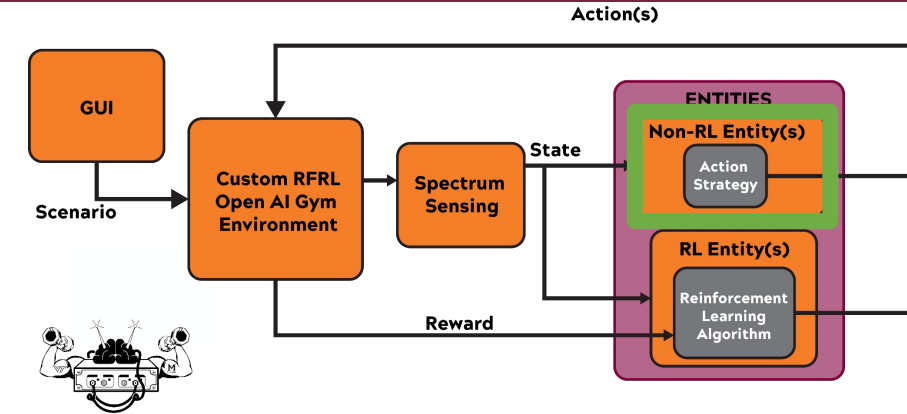
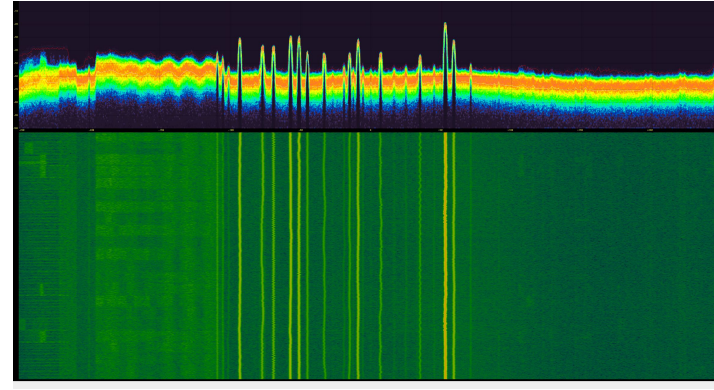
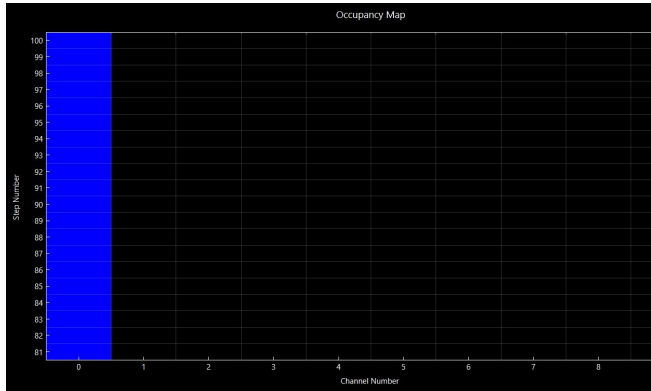


$$X(f, \tau) = \frac{1}{N} \sum_{t=1}^N x(t) w(t - \tau) e^{-2\pi j f t / N}$$

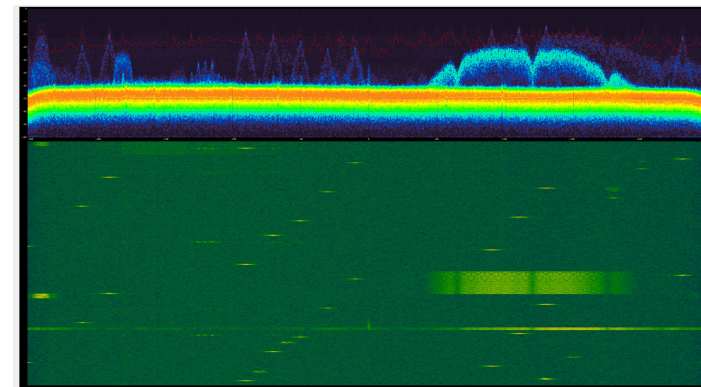
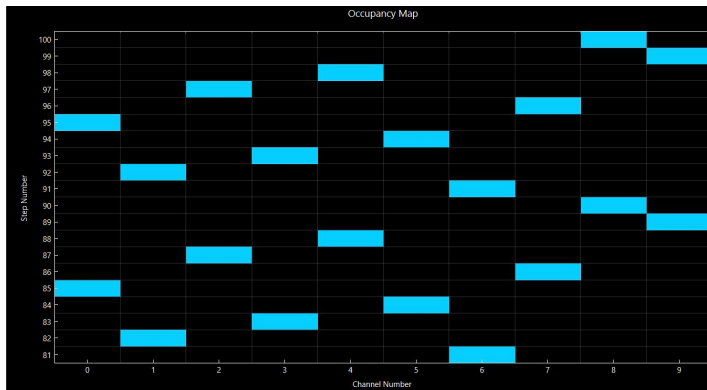
$$E[x_1] = \frac{1}{K} \sum_{i=1}^K |x_1[i]|^2$$

Entities: Non RL Control

Constant Frequency Entity: Stays in one channel



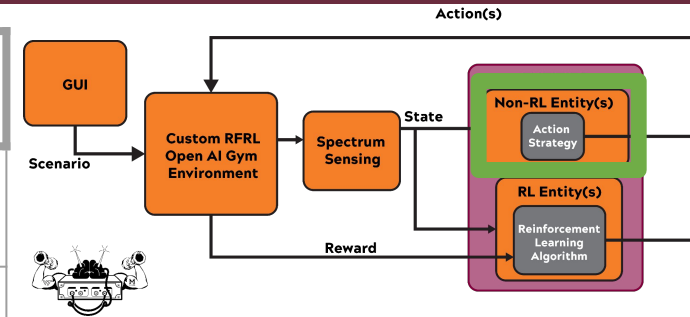
Hopper Entity: Move around with user-defined pattern



All Entities

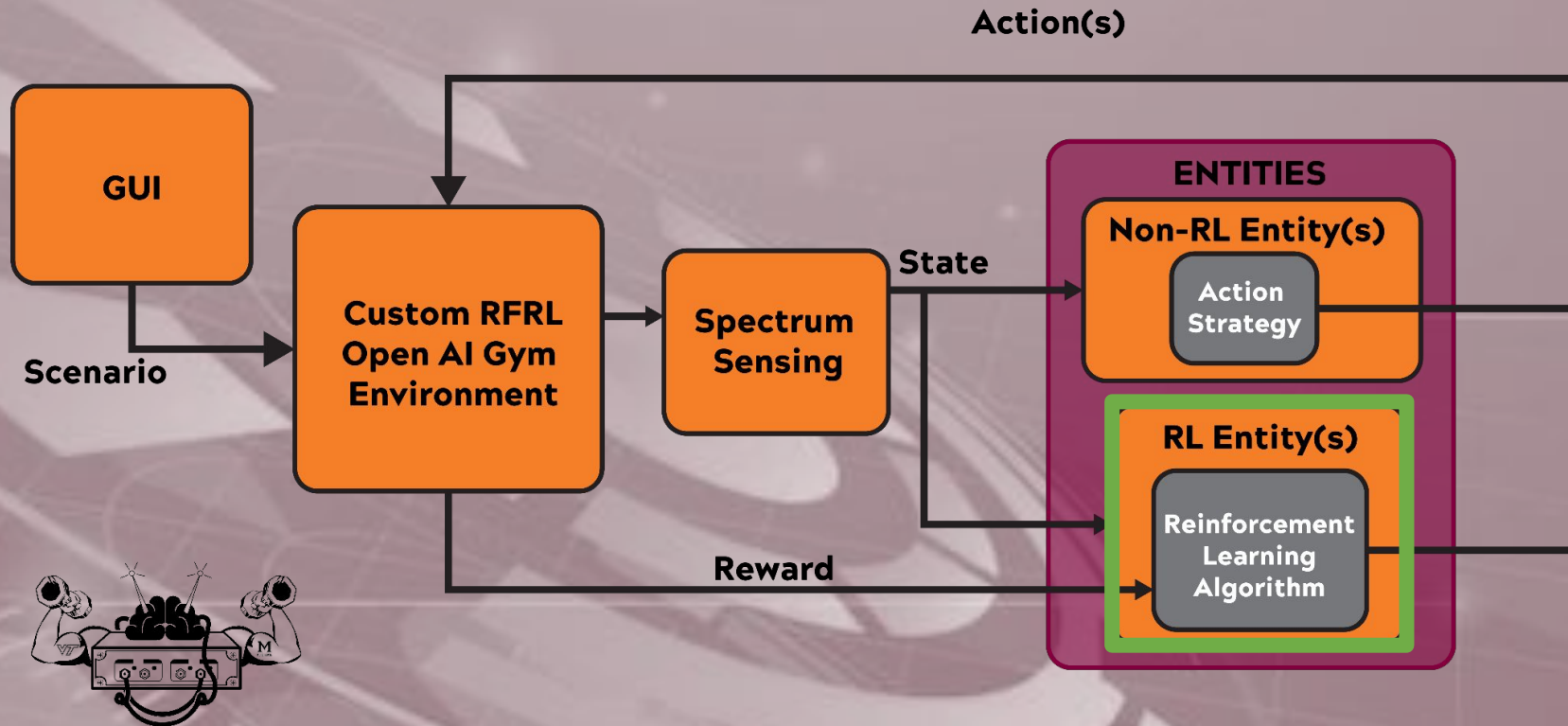


Entity Type	Description	Real World Example
Constant	Remains in one channel	T.V. or Radio
Fixed Hopper	Moves in user-defined pattern	Bluetooth
Stochastic Hopper	Selects channel based on user-defined probabilities for each channel	Multi-Armed Bandit
Agile Hopper	Moves to empty channel if available. Has perfect knowledge of other entities in environment	CR with DSA
Simple Jammer	Moves to an occupied channel. Has perfect knowledge of other entities in environment	Intelligent Adversary
Custom	Up to the user!	





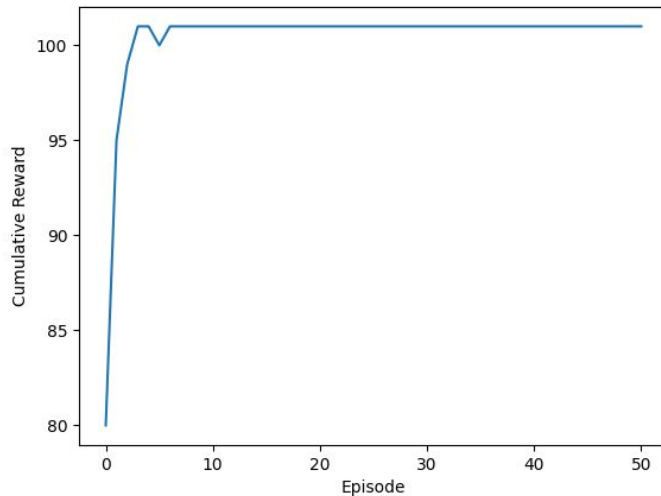
Validation and Analysis



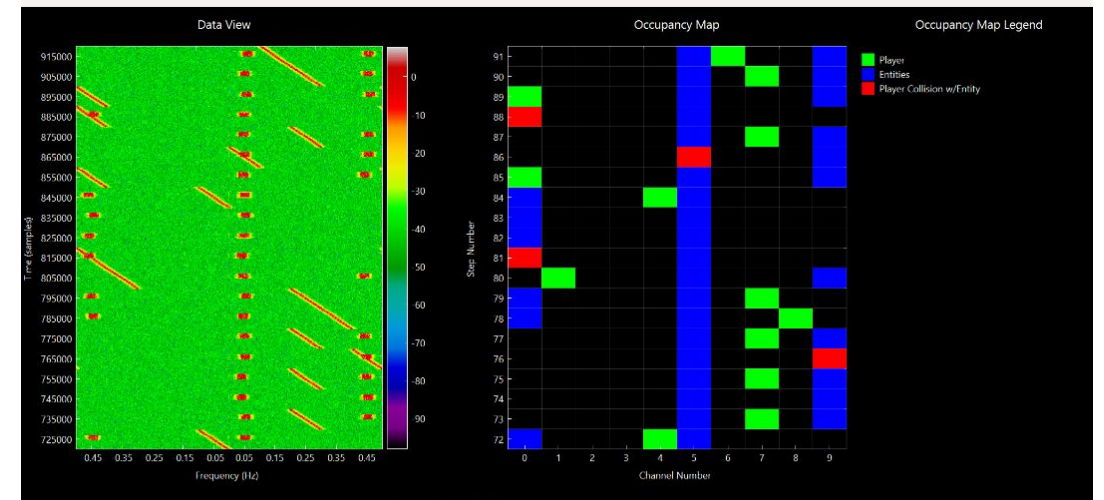
Scenario: Solvable Multi-Entity DSA

Scenario: 1 Constant Freq, 1 Stochastic Hopper

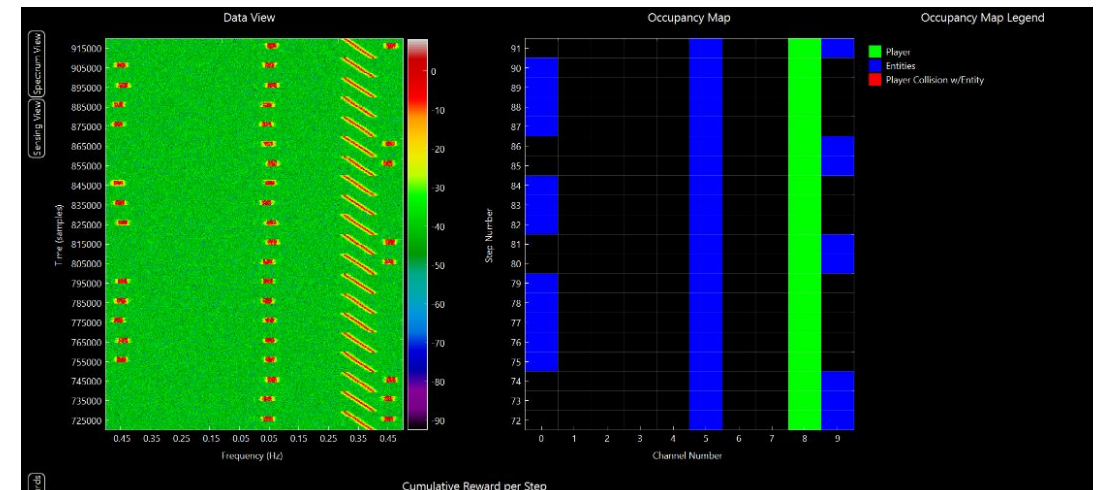
$$DSA \text{ Reward} = \begin{cases} 1 & \text{no collision} \\ -1 & \text{collision} \end{cases}$$



Before Convergence

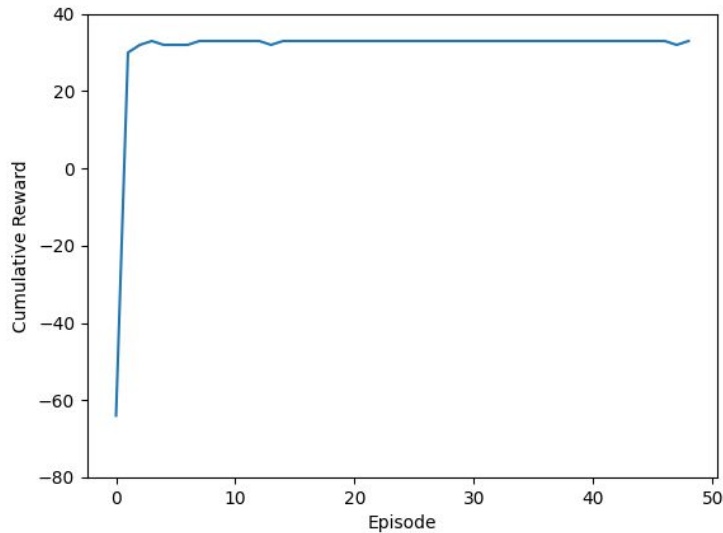
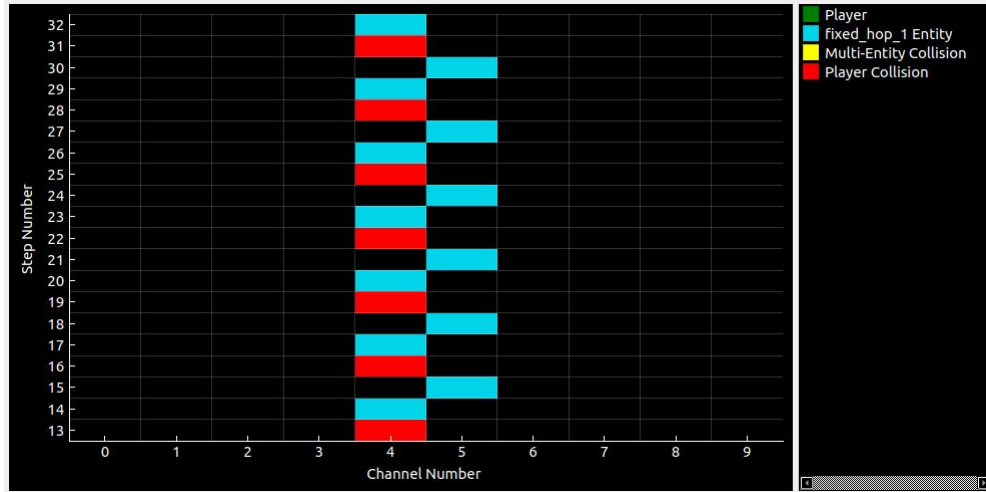


After Convergence



Findings from Challenging Scenarios

Scenario: Non-Markovian Jamming

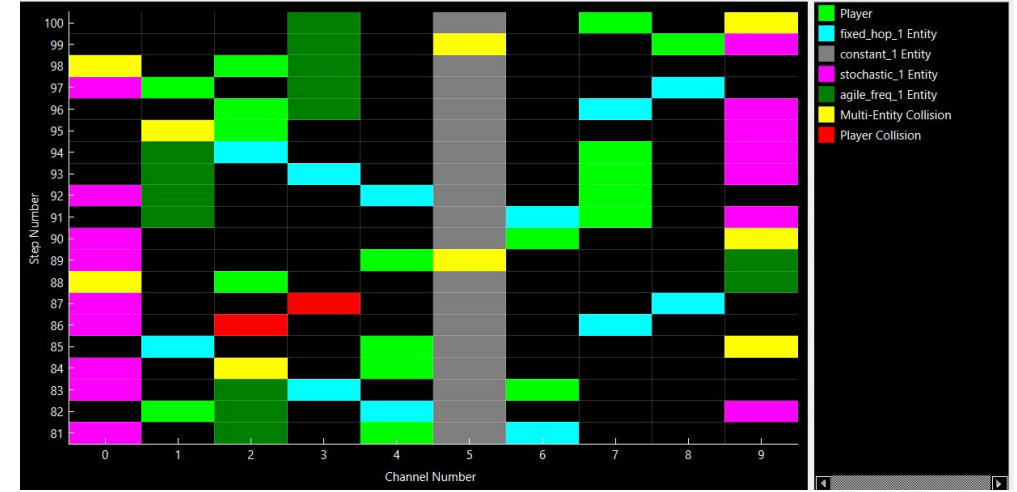


Suboptimal
Convergence

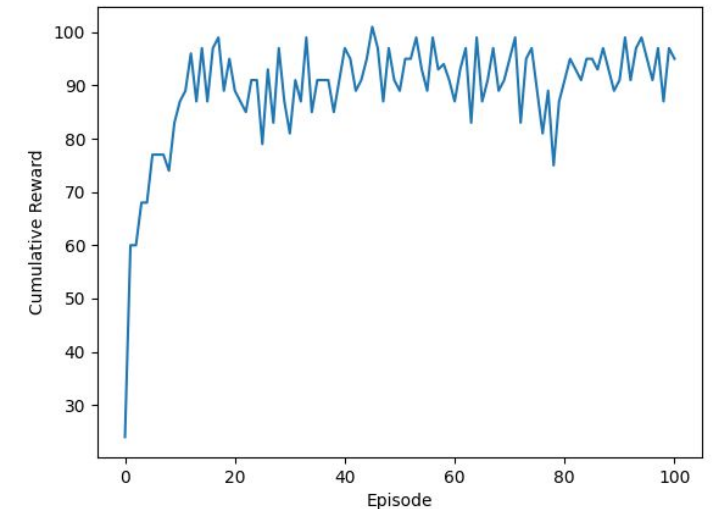
$$100 \left(\frac{1}{3}(1) + \frac{2}{3}(0) \right) = 33$$

The RFRL Gym can show the
limitations of Algorithms!

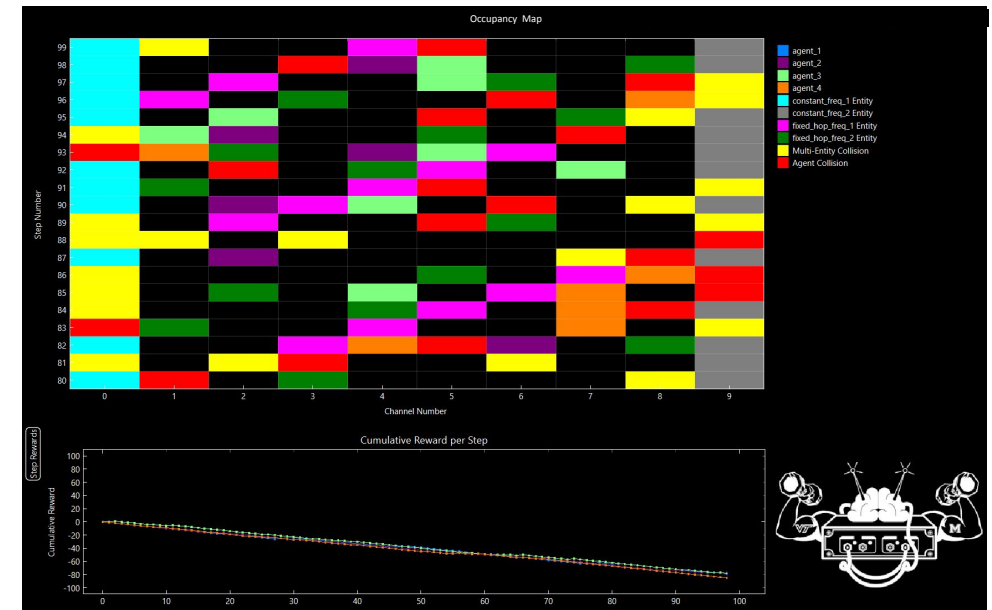
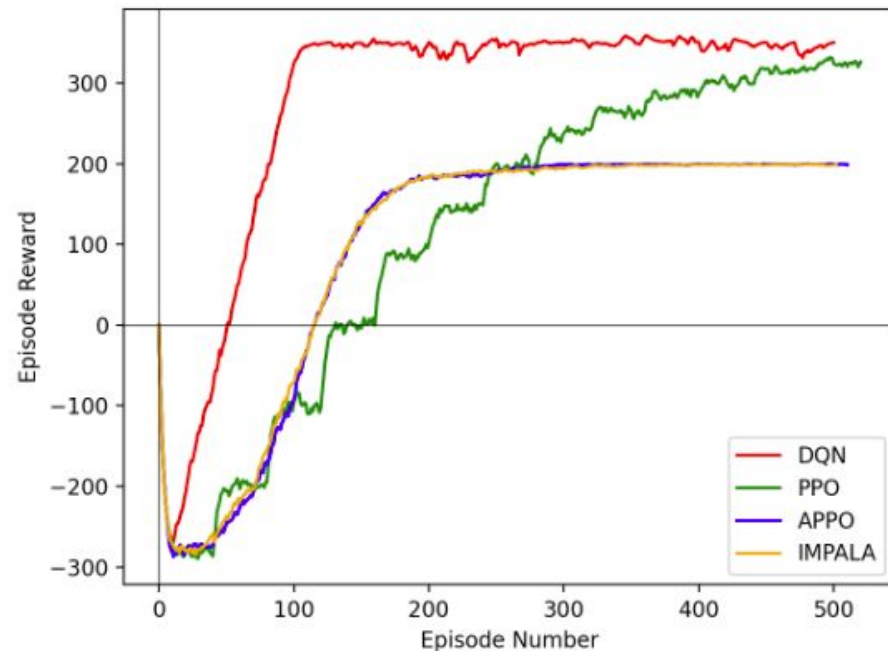
Scenario: Non-Deterministic Entity DSA



Noisy
Rewards



- Hardware Integration
- Expansion on observation spaces: modulation classification
- GUI Expansion: end to end user interaction
- Multi-Agent Reinforcement Learning (Paper in Progress)





Codebase: <https://github.com/vtnsiSDD/rfml-gym>
daniel20@vt.edu, alysemjones@vt.edu, cheadley@vt.edu



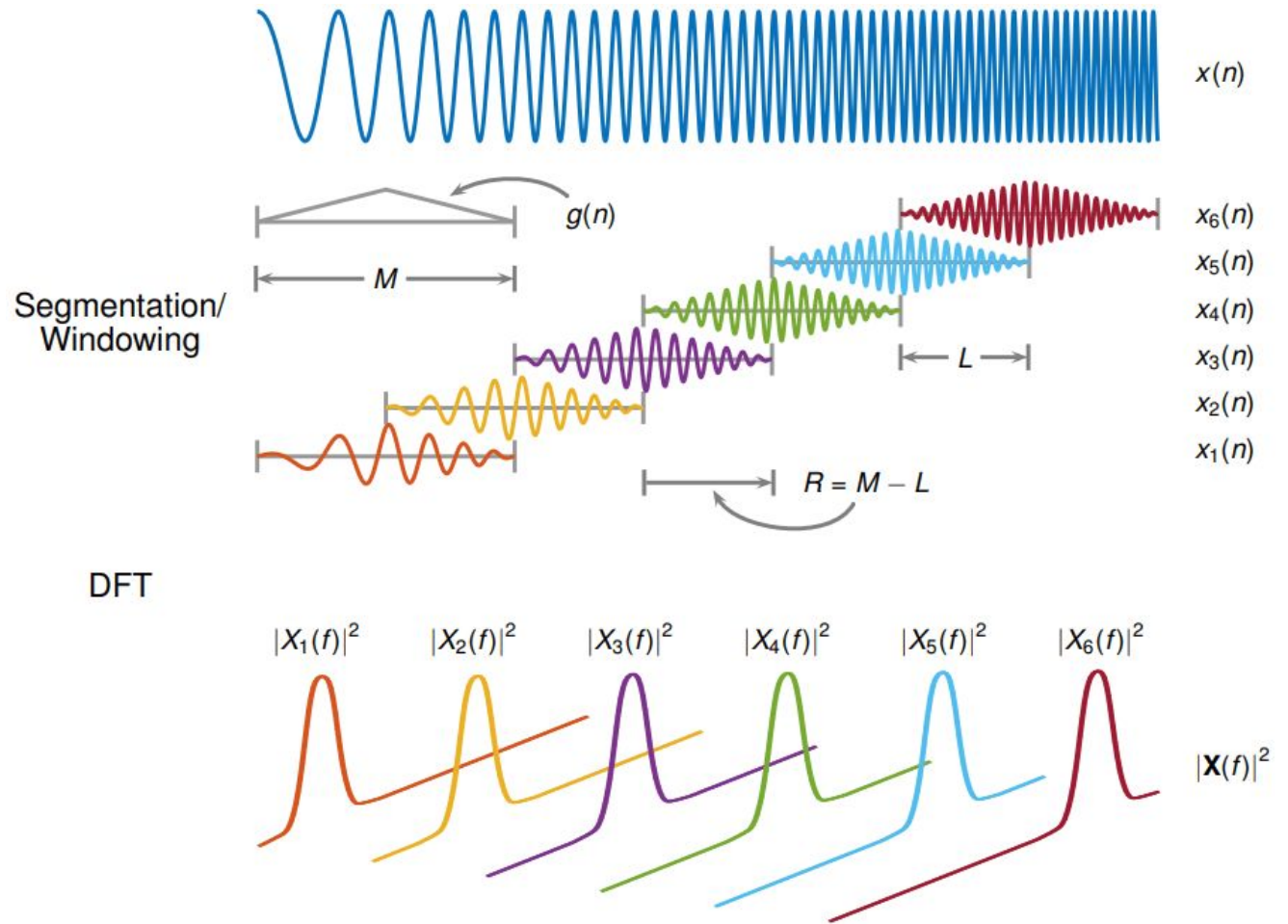
Thank You
Questions?

- [1]“RF Solutions,” www.maxar.com. <https://maxar.com/products/rf-solutions> (accessed Dec. 13, 2023).
- [2]“Summary of ctia’s annual wireless industry survey,”
<https://api.ctia.org/wp-content/uploads/2022/09/Summary-of-CTIAs-Wireless-Industry-Survey-2022.pdf>
- [3]M. Reed, “FCC Spectrum Decision Good for App Makers,” ACT | The App Association.
<https://actonline.org/2014/05/15/todays-fcc-spectrum-decision-great-for-app-makers/> (accessed Dec. 13, 2023).
- [4]Selvaraj, Janani. (2016). INVESTIGATIVE STUDY ON ROUTING METHODS FOR CRSN. International Journal of Latest Trends in Engineering and Technology. 10. 51-60. 10.21172/1.102.08.
- [5]D. Rodrigo, Pedro Ponce Cruz, and A. Molina, “CODA Algorithm: An Immune Algorithm for Reinforcement Learning Tasks,” InTech eBooks, Sep. 2016, doi: <https://doi.org/10.5772/63570>.
- [6]“Phase Modulation of Atomic Signal,” NIST, Accessed: Dec. 13, 2023. [Online]. Available:
<https://www.nist.gov/image/phase-modulationjpg-0>
- [7] Miah, Md Sipon & Yu, Heejung & Godder, Tapan & Rahman, M Mahbubur. (2015). A Cluster-Based Cooperative Spectrum Sensing in Cognitive Radio Network Using Eigenvalue Detection Technique with Superposition Approach. International Journal of Distributed Sensor Networks. 2015. 10.1155/2015/207935.
- [8] P. Pandya, A. Durvesh and N. Parekh, "Energy Detection Based Spectrum Sensing for Cognitive Radio Network," *2015 Fifth International Conference on Communication Systems and Network Technologies*, Gwalior, India, 2015, pp. 201-206, doi: 10.1109/CSNT.2015.264.

Spectrogram

A series of overlapped FFTs to illustrate how the frequency domain changes over time

$$X(f, \tau) = \frac{1}{N} \sum_{t=1}^N x(t) w(t - \tau) e^{-2\pi j f t / N}$$



Acquired from <https://www.mathworks.com/help/signal/ref/stft.html>

Spectrum Sensing Detector - PSD-based Detector

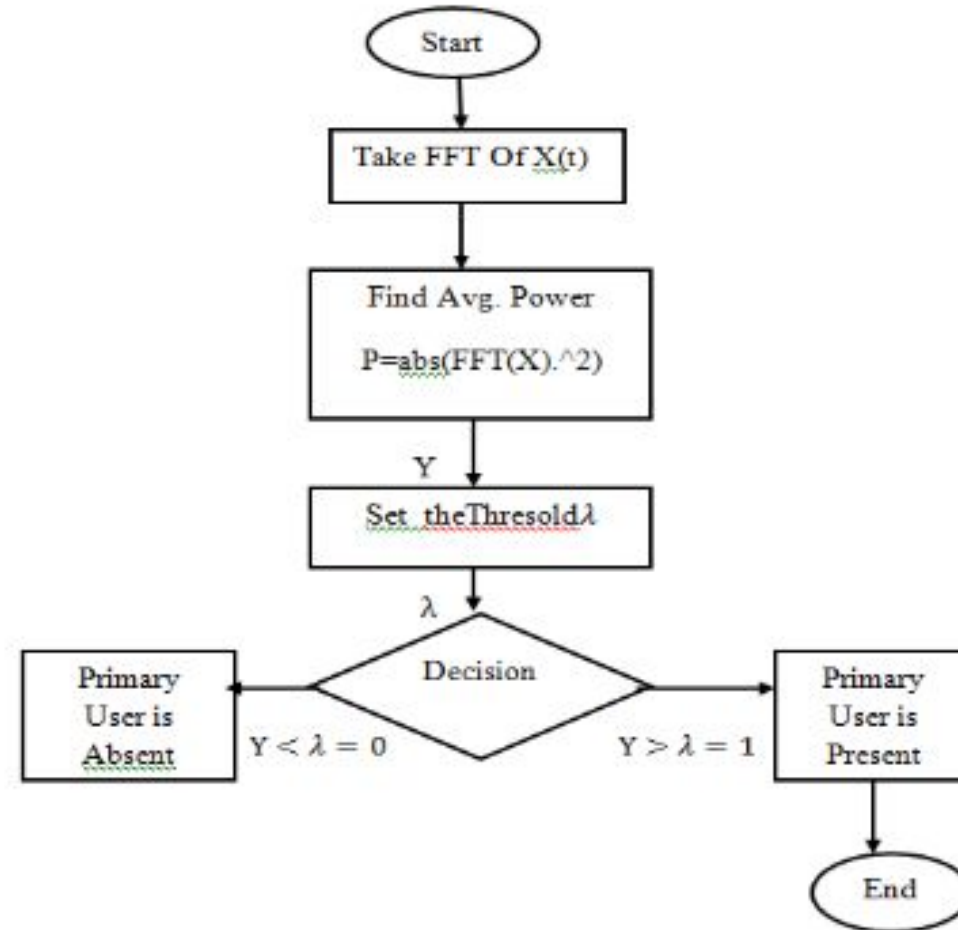
- Splits the wideband spectrum into N sub-channels
 - Ex: 1 MHz wide band sample rate, $N = 8$ filters, each sub-channel 125 kHz wide
- Each output represents the *state* of the RF spectrum
- After split into N sub-channels, the energy of one sub-channel is calculated

$$E[x_1] = \frac{1}{K} \sum_{i=1}^K |x_1[i]|^2$$

- Hypothesis test performed to distinguish signal energy from the noise floor

H_0	$y[n] = w[n]$	$E[x_1] < \lambda$	Unoccupied (0)
H_1	$y[n] = x[n] + w[n]$	$E[x_1] > \lambda$	Occupied (1)

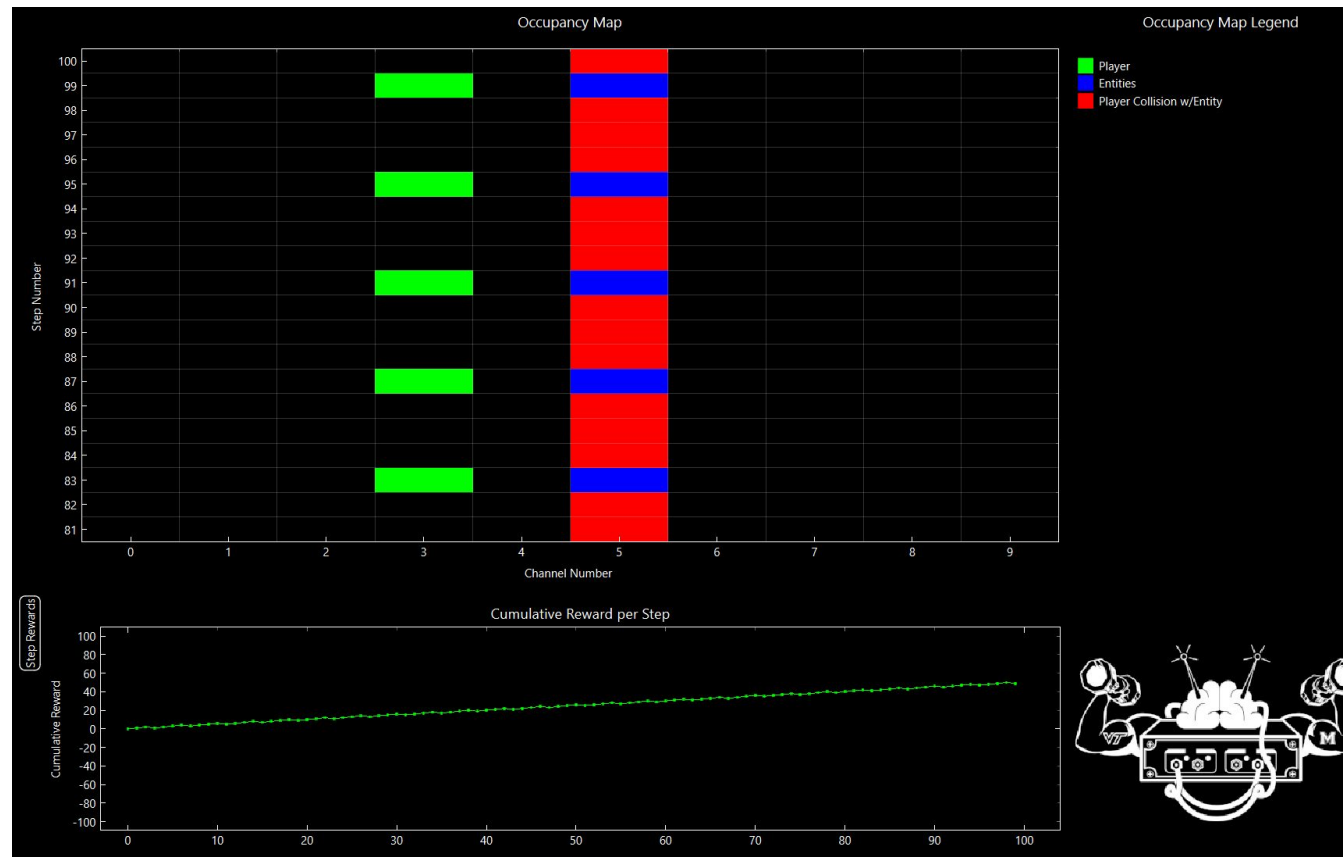
Energy Detection Algorithm

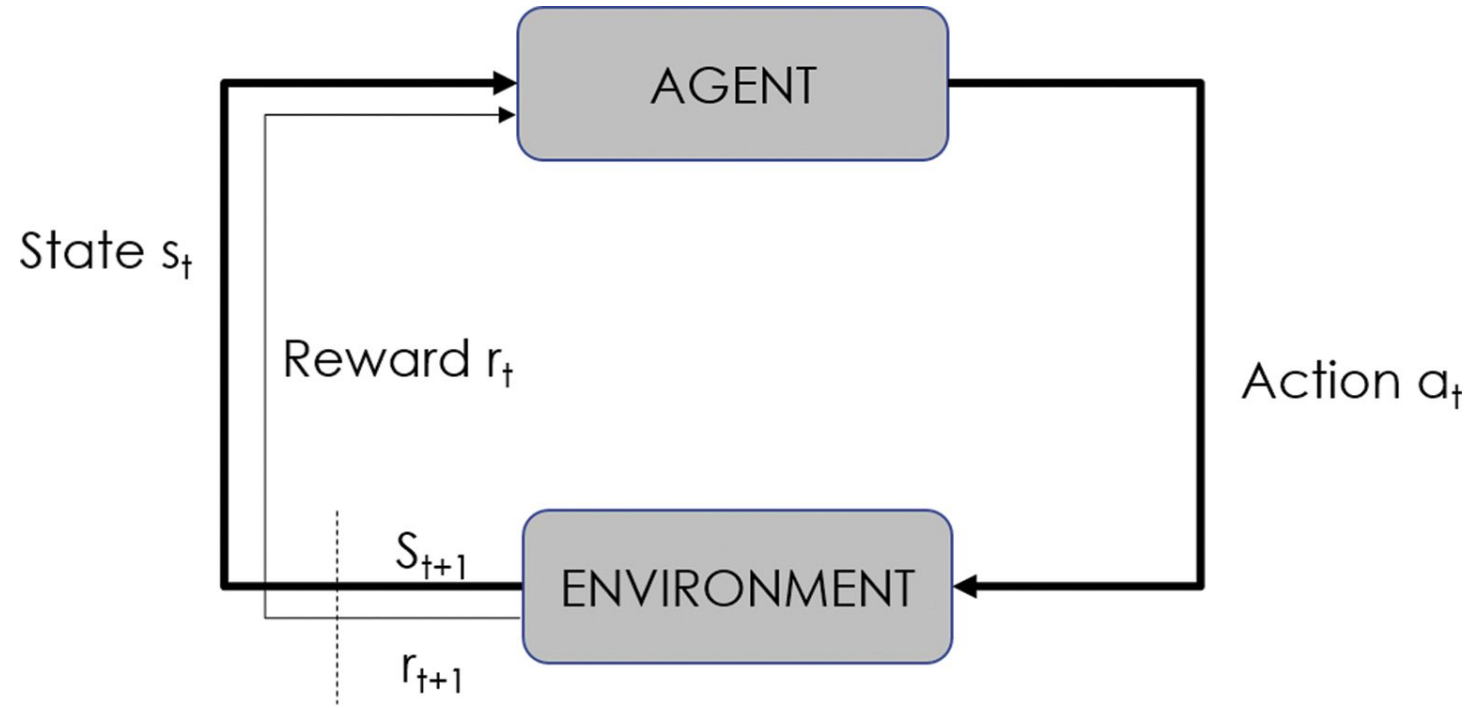


Jamming Scenario

Scenario: 1 Constant Freq

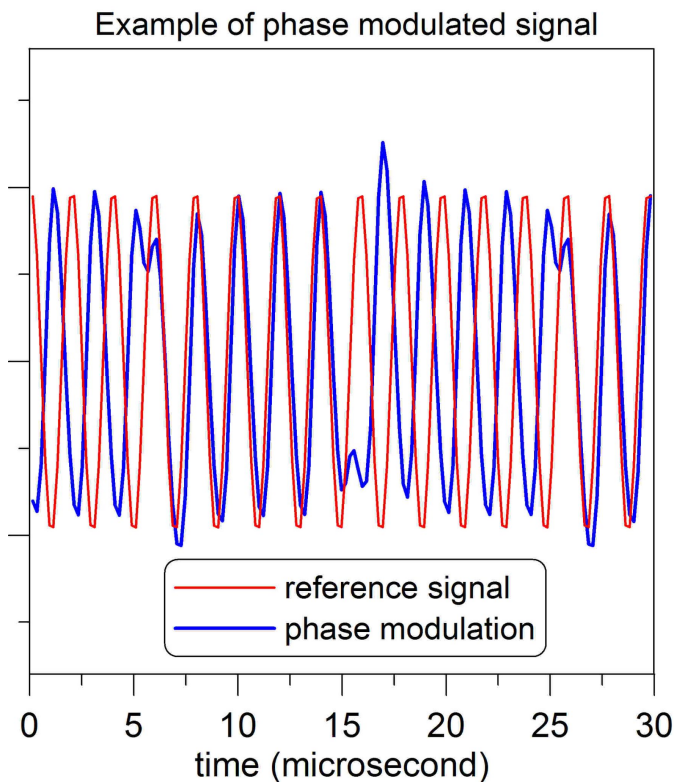
$$\text{Jamming reward} = \begin{cases} 1, & \text{collision} \\ -1, & \text{no collision} \end{cases}$$





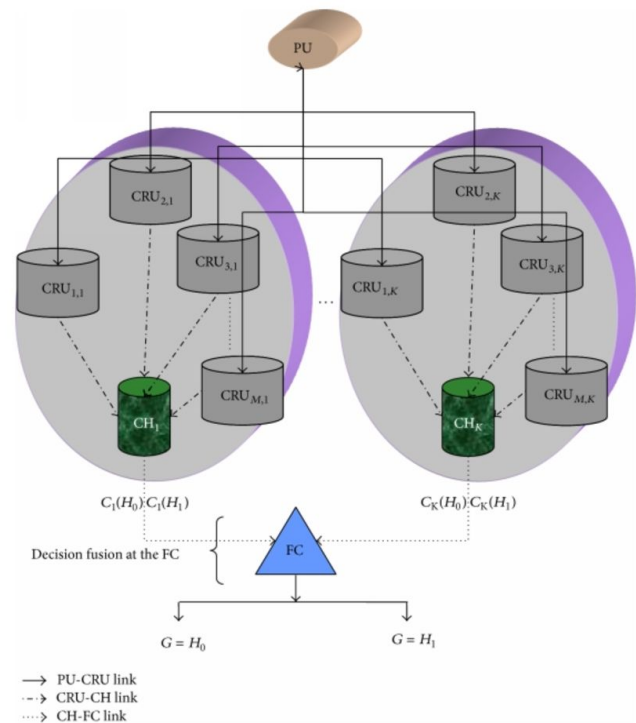
Acquired from [5]

Signal Modulation



[6]

Collaborative Spectrum Sensing



[7]

Modulation Classification

